Punit Singh

AI Research Engineer | Bihar, India

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PROFESSIONAL SUMMARY

AI Research Engineer with deep expertise in NLP and transformer architectures, fine-tuning LLMs for enterprise applications. Published internal research on model optimisation techniques. Passionate about advancing applied AI within the Indian tech ecosystem.

TECHNICAL SKILLS

MLOps & Deployment: MLOps, MLflow, Model Deployment

Data Engineering: ETL, Hadoop

Programming Languages: Python, SQL, C++, R, Java

ML Libraries & Frameworks: XGBoost, LightGBM

Soft Skills: Analytical Thinking, Project Management, Adaptability, Team Collaboration, Critical Thinking

PROJECTS

Real-Time Object Detection for Smart Surveillance

Tech Stack: Python

- Trained a YOLOv8 model on a custom 15K-image dataset for multi-class object detection.
- Achieved mAP@50 of 0.87 with real-time inference at 30 FPS on edge devices.
- Integrated OpenCV pipeline for video stream processing and alert generation.

Medical Image Classification using CNN

Tech Stack: Python

- Built a ResNet-50 transfer learning model for pneumonia detection from chest X-rays.
- Trained on NIH dataset (10K images), achieving 94% accuracy and 0.92 AUC.
- Implemented Grad-CAM heatmaps for model interpretability and clinical validation.

Predictive Churn Model for Telecom

Tech Stack: Python, MLflow

- Engineered 45+ features from raw telecom usage data including RFM and behaviour signals.
- Compared XGBoost, LightGBM, and Random Forest; selected LightGBM at 88% AUC-ROC.
- Tracked 30+ experiments using MLflow and registered best model to the model registry.

EDUCATION

B.Tech – Data Science

National Institute of Technology (NIT) Agartala | 2025 | CGPA: 9.38

Class XII – CBSE

2020 | 75.4%

Class X – CBSE

2018 | 79.8%